

Computer Vision Guidelines

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1.1 Derivative in Computer Vision

Image derivatives are the mathematical foundation of both edge and corner detection, as they capture local intensity variations that correspond to meaningful structural features. A 2D image can be treated as 2D discrete function $I(x, y)$. The derivatives are approximated via finite differences or convolution with derivative kernels.

The first order derivatives explains the rate of change of intensity in an image, representing the edges as local maximum $\nabla I(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y})$. The second order derivative describes the rate of change of the gradient, and the edges are represented as zero crossing point (as in LoG) $\nabla^2 I(\frac{\partial^2 I}{\partial x^2}, \frac{\partial^2 I}{\partial y^2})$. However the LoG function is high computational complex, and it is approximated by DoG for computational efficiency.

1.2 Gradient magnitude and direction

From the first-order derivatives I_x and I_y , — typically computed via the Sobel operator, a separable filter combining a derivative kernel with a smoothing kernel — two quantities are derived:

- Magnitude: $|\nabla I| = \sqrt{I_x^2 + I_y^2}$, which encodes the edge strenght (high values represents a sharp transition)
- Direction: $\theta = \arctan(\frac{I_y}{I_x})$, which encodes the edge orientation (perpendicular to edge contours)

These quantities are crucial to apply NMS, since the edge direction is used to determine the orientation of the local neighborhood for non-maximum suppression, ensuring that only the most significant edge pixels are retained and edge magnitude is used to threshold the edges, discarding weak edges that are likely to be noise.

1.3 Second order derivative for Intensity transitions

The laplacian (LoG) responds to intensity transitions with a zero corssing at the edge location, remarking edge boundary with high spatial accuracy. However, its main challenge is noise since it is amplified by the function, solved by a pre-smoothing by a Gaussian filter. Two main strategies are used to overcome the noise sensitivity:

- Gaussian smoothing, before differentiation
- Medial filtering, non linear strategy for salt and pepper noise

1.4 Corner detection

Corner detection relies on image derivatives and a second-moment matrix approach (structure tensor). The second moment matrix H over a window W is defined as:

$$H = \sum_{(x,y) \in W} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

To analyze the local intensity variation, we study the eigenvalues λ_1 and λ_2 of H , which correspond to the magnitudes of maximum and minimum intensity change:

- $\lambda_1 \approx \lambda_2 \approx 0$ (Flat region): The error is low in all directions; intensity is nearly constant.
- $\lambda_1 \gg \lambda_2 \approx 0$ (Edge): The error is high in one direction but low along the edge.
- $\lambda_1 \approx \lambda_2 \gg 0$ (Corner): The error is high in all directions, representing a point of distinct intersection.

Since calculating eigenvalues for every pixel is computationally expensive, the Harris detector introduces a response function R that approximates their behavior without explicit eigenvalue decomposition:

$$R = \det(H) - k \cdot (\text{trace}(H))^2 = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$$

where k is an empirical constant (typically 0.04 – 0.06). The value of R indicates the "cornerness":

- R large and positive $\implies$ Corner.
- R large and negative $\implies$ Edge.
- $|R|$ small $\implies$ Flat region.

Properties of the Harris detector:

- **Equivariant** to translation and rotation (eigenvalues and shape are preserved under rotation).
- **Invariant** to additive brightness shifts ($I' = I + b$) because derivatives eliminate constant offsets.
- **Not scale-invariant**, as a physical corner may appear as an edge or a flat region when zoomed in.



2.1 Detecting keyframes in Video

The all approach can be divided in 3 main features, to detect all the different situation in a video sequence.

- Macro (global changes)

- Micro changes
- Optical flow magnitude for movements

The all algorithm is completed by using a minimum gap constraint to avoid considering a number k of successive frames, to not face the redundancy problem.

2.2 Macro

To have a general and comprehensive understanding of the scene, the color histogram in HSV can be analyzed. This approach is not complex and allows to detect scene interrupts and macro changes in the video sequence. The two histograms are compared with Chi square distance.

2.3 Micro

It may happen that the color histogram in two consequent scenes is similar, but the objects displacement is completely different. This issue may be overcome by the use of an edge ratio system: a canny edge detector algorithm is used to find all edges in a scene and the amount of edges in two scenes is compared finding a change ratio.

2.4 Optical Flow

Finally, to detect the beginning of actions, the optical flow magnitude can be used $M = \frac{1}{N} \sum \sqrt{u^2 + v^2}$. LK comparing algorithm is used to compare consecutive frames and it can detect in two scenes some movements (mouth moving while talking), where the scenes appear similar to the Macro analyzer.



3.1 Padding Strategies

Padding is used to allow the filter to shift on all pixels of the image, including corners and edges. Many different strategies of padding exist:

- Zero and Constant padding: The borders are filled with 0s or constants, creating artificial edges on the borders

- **Mirror:** Reflects the inner pixel values excluding the border pixels
- **Symmetric:** Reflects the inner pixel values including border pixels
- **Clamp:** Replicates the most external pixel constantly

3.2 Effect on Output Dimensions (Valid vs. Same vs. Full)

The size of the output image depends on the input size N , kernel size K , padding size P , and stride S :

$$\text{dim}_{\text{out}} = \left\lfloor \frac{N + 2P - K}{S} \right\rfloor + 1$$

- **Valid** ($P = 0$): No padding is applied. The output is smaller than the input ($\text{dim}_{\text{out}} = N - K + 1$ for $S = 1$).
- **Same** ($P = \lfloor K/2 \rfloor$): Padding is chosen such that the output has the exact same spatial dimensions as the input (for $S = 1$).
- **Full** ($P = K - 1$): Padding is added such that every element of the kernel visits every input pixel. The output is larger than the input ($\text{dim}_{\text{out}} = N + K - 1$).

3.3 Impact on Edge Detection Near Boundaries

Edge detection relies on computing spatial derivatives (gradients):

- **Zero and Constant Padding** ($c \neq 0$): Introduce a sharp, artificial step-discontinuity between real pixel intensities and the constant padding value. Derivative filters (like Sobel) mistake this jump for a strong edge, generating **false/spurious edges** along the image borders.
- **Clamp / Replicate Padding:** Forces the gradient at the boundary to zero by replicating pixel values. While it avoids artificial step-edges, it incorrectly ****suppresses legitimate real edges**** crossing near the boundary.
- **Mirror and Symmetric Padding:** Preserve smooth gradient continuity without sharp steps. They are the optimal strategies for edge detection near image boundaries as they ****correctly preserve edge responses**** without creating false positives or suppressing true borders.



4.1 Estimating optical flow

To estimate optical flow, the Lk algorithm is employed. It is based on three main assumption:

- Brightness constancy: $I(x, y, t) = I(x + u\delta t, y + v\delta t, t + \delta t)$
- Small motion
- Spatial coherence

Starting from the brightness constancy, by applying Taylor approximation, we can derive the optical flow constraint equation $I_x u + I_y v + I_t = 0$. Since this single-pixel equation is underconstrained (1 equation for 2 unknowns), Lucas-Kanade introduces the spatial coherence assumption, yielding N^2 equations for an $N \times N$ patch. $Ax = b$ is so formed as an overdetermined system solvable by Least Squares. The Solution is the eigenvector corresponding to the smallest eigenvalue. This is possible only if $A^T A$ is invertible and well conditioned. Indeed, by the $A^T A$ eigenvalues we can understand the optical flow, as for the harris corner detector:

- $\lambda_1 \approx \lambda_2 \approx 0$ (Flat region): No reliable optical flow can be estimated.
- $\lambda_1 \lambda_2 \approx 0$ (Edge): Aperture problem; only the normal flow (perpendicular to the edge) can be estimated.
- $\lambda_1 \approx \lambda_2 \gg 0$ (Corner / Texture): Optical flow can be estimated reliably in all directions.

The ****aperture problem**** occurs in the edge case, stating that observing a moving edge through a small local aperture only allows estimating the motion component perpendicular to the edge, while the parallel (tangent) component remains completely ambiguous. Lucas-Kanade mitigates this by pooling information from a spatial neighborhood (patches) rather than relying on isolated pixels.



5.1 Role of DFT in Image Filtering

The Discrete Fourier Transform (DFT) converts an image from the spatial domain into the frequency domain, decomposing it into a weighted sum of 2D sinusoids. Its primary role in image filtering relies on the **Convolution Theorem**, which states that a spatial convolution $g * h$ is equivalent to an element-wise multiplication in the frequency domain:

$$\mathcal{F}\{g * h\} = \mathcal{F}\{g\} \cdot \mathcal{F}\{h\}$$

Instead of performing costly sliding-window spatial convolutions—whose computational complexity grows quadratically with the kernel size as $O(L^2N^2)$ —filtering can be efficiently executed in $O(N^2 \log N)$ using the Fast Fourier Transform (FFT). By the Convolution Theorem, this global approach completely eliminates the need for local window movement: we transform the entire image via FFT, transform the kernel—**zero-padded to the image dimensions to avoid circular convolution artifacts**—via FFT, perform a single global element-wise multiplication of their spectra, and apply the **Inverse FFT (IFFT)** to return to the spatial domain. This method is especially advantageous when applying very large filter kernels.

5.2 Information Provided by the DFT

The DFT transforms an image into a complex-valued 2D spectrum $F(u, v) = R(u, v) + iI(u, v)$, providing two fundamental types of information:

- **Amplitude Spectrum (Magnitude):** $|F(u, v)| = \sqrt{R^2 + I^2}$, representing *how much energy* of a specific spatial frequency is present in the image.
 - Low frequencies (near the center, DC component) correspond to smooth, large-scale structures and background.
 - High frequencies (far from the center) correspond to fine details, sharp edges, and rapid intensity variations (textures).
- **Phase Spectrum:** $\phi(u, v) = \text{atan2}(I, R)$, which encodes the *spatial alignment* of those frequency components, defining the exact geometric location of edges, lines, and structures in the image space. **Phase is perceptually more important than amplitude:** in phase-swapping experiments, reconstructing an

image using the amplitude of Image A and the phase of Image B yields an image whose structural layout and recognizable shapes belong strictly to Image B.

5.3 Filtering Examples: Low-Pass and High-Pass

- **Low-Pass Filtering:** Used for smoothing or blurring. We multiply the frequency spectrum by a mask (such as an ideal cutoff mask beyond a radius r or a smooth 2D Gaussian function) that preserves low frequencies near the center and attenuates high frequencies. *Example:* Removing high-frequency Gaussian sensor noise or smoothing background details.
- **High-Pass Filtering:** Used for sharpening or edge detection. We multiply the frequency spectrum by a mask that attenuates low frequencies at the center and preserves high frequencies in the outer peripheral regions. *Example:* Isolating abrupt intensity transitions to highlight edges and fine textures.



6.1 SIFT Detection

The Scale-Invariant Feature Transform (SIFT) detects stable keypoints in scale-space by constructing a **Difference of Gaussians (DoG)** pyramid across octaves:

$$\text{DoG}(x, y, \sigma) = G(x, y, k\sigma) - G(x, y, \sigma)$$

A pixel is selected as a keypoint if it is a local extremum compared to its **26 neighbors** in the 3D scale-space (8 spatial, 9 in the scale above, 9 below). Unstable points—specifically low-contrast points and edge responses (filtered via a 2D Hessian ratio test)—are discarded. Finally, a **canonical orientation** is assigned based on the peak of a local gradient orientation histogram to ensure rotation invariance.

6.2 SIFT Descriptor

To represent the local appearance around a keypoint:

- A 16×16 pixel window is extracted at the keypoint's characteristic scale.
- The coordinate system and gradients are rotated according to the keypoint's canonical orientation (achieving rotation invariance).

- The window is divided into a 4×4 grid of cells (16 cells total).
- An **8-bin orientation histogram** is computed for each cell.
- Concatenating all 16 histograms yields a single **128-dimensional vector** ($16 \times 8 = 128$).
- The vector is normalized to unit length (L_2 norm) for illumination invariance.

6.3 SIFT Properties

The SIFT algorithm is robust and invariant to:

- **Scale:** Via multi-scale DoG extrema detection.
- **Rotation:** Via canonical orientation alignment.
- **Illuminazione:** Via L_2 vector normalization.



7.1 SURF Detection

SURF (Speeded-Up Robust Features) is a faster alternative to SIFT, designed to address high computational complexity while maintaining comparable matching quality.

Instead of using Difference of Gaussians (DoG), SURF is a **Hessian-based** detector that identifies blobs at locations where the determinant of the Hessian matrix H is maximal:

$$H(x, \sigma) = \begin{bmatrix} L_{xx}(x, \sigma) & L_{xy}(x, \sigma) \\ L_{xy}(x, \sigma) & L_{yy}(x, \sigma) \end{bmatrix}$$

where L_{xx} represents the convolution of the image with the second partial derivative of a Gaussian.

To achieve high computation speed, SURF approximates these Gaussian second-order derivatives using **box filters**, which can be evaluated extremely fast in constant time using **integral images** (where each location stores the sum of all pixel intensities in the rectangle from the origin). The approximate determinant is computed as:

$$\det(H_{\text{approx}}) = D_{xx}D_{yy} - (wD_{xy})^2 \quad (w = 0.9)$$

7.2 SURF Descriptor

The SURF descriptor is designed to be compact and fast to compute:

- **Orientation assignment:** Computes Haar wavelet responses in the x and y directions within a circular neighborhood around the interest point, weighted by a Gaussian, to find a dominant orientation (ensuring rotation invariance).
- **Feature vector:** A square window of size $20s \times 20s$ (where s is the scale) is aligned with the dominant orientation and divided into a 4×4 grid of cells (16 cells total).
- For each cell, the sum of Haar wavelet responses and their absolute values are computed: $\sum dx, \sum dy, \sum |dx|, \sum |dy|$.
- Concatenating these 4 values for all 16 cells creates a **64-dimensional descriptor vector** ($16 \times 4 = 64$), making it computationally lighter and more compact than SIFT's 128-D vector.



8.1 Camera Model and Projection Matrix

Recovering 3D structure from images requires modeling how 3D world points are mapped to 2D pixel coordinates. This is expressed by the 3×4 projection matrix $P = K[R \mid t]$, which unifies the intrinsic and extrinsic parameters.

8.2 Intrinsic and Extrinsic Parameters

- **Intrinsic Matrix (K):** Encodes internal optical properties, mapping 3D camera coordinates to 2D pixels using focal lengths (f_x, f_y) and the principal point (o_x, o_y):

$$K = \begin{bmatrix} f_x & 0 & o_x \\ 0 & f_y & o_y \\ 0 & 0 & 1 \end{bmatrix}$$

- **Extrinsic Matrix ($[R \mid t]$):** Describes the camera's pose relative to the world frame, consisting of a 3×3 rotation matrix R and a 3×1 translation vector t .

8.3 Camera Calibration

Camera calibration consists of computing the transformation parameters by estimating the 3×4 matrix P . Using at least **6 point correspondences** between known 3D world points and 2D image points (since P has 11 degrees of freedom), we obtain 2 linear equations per point. This forms a homogeneous system $A\mathbf{p} = 0$, solved via SVD by finding the eigenvector corresponding to the smallest eigenvalue of $A^T A$.

Once $P = [M \mid p_4]$ is estimated, intrinsic and extrinsic parameters are decoupled using **RQ factorization**:

$$M = KR \implies R = K^{-1}M, \quad p_4 = Kt \implies t = K^{-1}p_4$$

8.4 Triangulation

Once the camera matrices (P, P') and corresponding 2D points (x, x') are known from two calibrated views, the 3D position of a scene point X is recovered via **triangulation**. By using the cross product to eliminate the unknown scale factor:

$$x \times (PX) = 0 \quad \text{and} \quad x' \times (P'X) = 0$$

This generates a homogeneous linear system $AX = 0$ (4×3 system), which is solved using SVD (minimizing the algebraic error) to estimate the 3D coordinates of the point. The solution considers also the noise challenge: it is possible that the two rays are skew, thus the solution involves the minimization of the consequent error.



9.1 Introduction to Epipolar Geometry and Epipoles

Epipolar geometry describes the geometric relationship between two camera views of the same 3D scene, providing constraints that simplify feature matching.

- **Baseline:** The line segment connecting the two camera optical centers (O and O').
- **Epipolar Plane:** The plane defined by a 3D point X and the two camera centers.

- **Epipoles (e, e'):** The intersection of the baseline with the image planes. e is the projection of O' onto the first image, and e' is the projection of O onto the second image. All epipolar lines converge at the epipole.
- **Epipolar Constraint:** Given a point x in the first view, its correspondence x' in the second view must lie on the corresponding epipolar line l' , reducing the search space from 2D to 1D.

9.2 Essential Matrix E and Fundamental Matrix F

- **Essential Matrix (E):** Encodes the epipolar constraint for **calibrated cameras** (K known), operating in normalized camera coordinates ($\mathbf{x}'^T E \mathbf{x} = 0$). It is defined as $E = R[t]_{\times}$, has 5 degrees of freedom (DoFs), and rank 2.
- **Fundamental Matrix (F):** The generalization of E for **uncalibrated cameras** (K unknown), operating directly in pixel coordinates ($\mathbf{p}'^T F \mathbf{p} = 0$). It maps pixels to epipolar lines ($l' = Fp$) and has 7 DoFs.
- **Algebraic Relationship:** F and E are linked through the intrinsic camera calibration matrices K and K' :

$$F = K'^{-T} E K^{-1} = K'^{-T} R[t]_{\times} K^{-1}$$

9.3 Estimation of the Fundamental Matrix

F is estimated from corresponding point matches using the **Normalized 8-Point Algorithm**:

1. **Normalization:** Translate and scale pixel coordinates (centroid at origin, average distance $\sqrt{2}$) to ensure numerical stability.
2. **Linear System:** Set up $Ah = 0$ using ≥ 8 point matches and solve via SVD (smallest singular value of $A^T A$).
3. **Rank-2 Constraint:** Enforce $\det(F) = 0$ by computing the SVD of the estimated matrix and zeroing out its smallest singular value ($\Sigma' = \text{diag}(\sigma_1, \sigma_2, 0)$).
4. **Denormalization:** Transform F back to the original coordinate system, typically combined with **RANSAC** to reject outliers.

The normalized algorithm is preferred since the normal algorithm is sensitive to noise and to avoid the poor numerical conditioning of the A matrix, which can result in high variance in the columns' values.

9.4 Relation to Calibration Parameters and Relative Poses

- **Calibration Parameters:** Intrinsic parameters (K, K') bridge pixel coordinates and normalized coordinates. Extrinsic parameters (R, t) define the relative rotation and translation between the two camera frames.
- **Recovering Relative Pose (R, t):** Once the Essential matrix E is recovered (either directly or via $E = K'^T F K$), it can be decomposed using SVD to yield four possible solutions for the relative camera pose (R, t). The correct physical solution is uniquely determined by checking that reconstructed 3D points lie in front of both cameras (cheirality constraint).

9.5 3D Triangulation

Given known camera projection matrices (P, P') and 2D correspondences (x, x'), the 3D position of a scene point X is recovered via **triangulation**. By eliminating the unknown scale factors via cross product ($x \times PX = 0$ and $x' \times P'X = 0$), we form a homogeneous system $AX = 0$ (4×3 system). Solving this via SVD minimizes the algebraic error to estimate the 3D spatial coordinates.



10.1 Definition of Disparity

Disparity d is the horizontal displacement (difference in pixel coordinates) of the same 3D point between a left stereo image and a right stereo image:

$$d = x_L - x_R$$

For objects in front of the cameras, $x_L > x_R$, yielding a positive disparity ($d \geq 0$). It is the core geometric parameter used to infer 3D depth.

10.2 Epipolar Rectification

Epipolar rectification is a preprocessing geometric transformation that warps both stereo images using homographies so that their epipolar lines become horizontal and coincide with the image scanlines ($v_L = v_R$). This simplifies the correspondence problem: instead of searching for matching points across a 2D image plane, the search is reduced to a 1D horizontal scan along the exact same row.

10.3 Block Matching and Similarity Metrics

To compute the disparity map, local window-based matching is commonly employed:

1. For each pixel in the left image I_L , a small patch (window) W is extracted.
2. The algorithm searches for the best matching patch along the same horizontal scanline in the right image I_R .
3. The optimal displacement d is selected using a **Winner-Takes-All (WTA)** strategy—choosing the disparity that minimizes the matching cost. Common dissimilarity metrics include:

- **SAD** (Sum of Absolute Differences):

$$\text{SAD} = \sum_{(u,v) \in W} |I_L(u, v) - I_R(u + d, v)|$$

- **SSD** (Sum of Squared Differences):

$$\text{SSD} = \sum_{(u,v) \in W} (I_L(u, v) - I_R(u + d, v))^2$$

- **NCC** (Normalized Cross-Correlation).

Window Size Trade-off: A small window captures fine details but results in a noisy disparity map; a larger window is robust to noise but blurs depth discontinuities and fails at localizing thin objects.

10.4 Relationship Between Disparity and Depth

Assuming a rectified stereo rig with two parallel cameras separated by a baseline b and sharing an effective focal length f , depth Z is inversely proportional to disparity d :

$$Z = \frac{b \cdot f}{d}$$

A high disparity indicates a close object (large horizontal shift), while a low disparity indicates a distant object (small shift).

10.5 Limitations and Failure Cases

Local block matching often fails in the following scenarios:

- **Textureless regions:** Homogeneous areas (e.g., a white wall) where all patches look identical, leading to severe matching ambiguity.
- **Repeated patterns:** Repetitive textures (e.g., bricks or fences) causing multiple identical minimum-cost matches.
- **Specular/reflective surfaces:** Mirrors or shiny metal violate the brightness constancy assumption.
- **Depth discontinuities:** Pixels near object boundaries where a local window covers parts of both the foreground and background.



11.1 Histogram-based vs. CNN-based Approaches

Consider two alternative approaches for segmenting a video into visually similar segments: **Approach A**, which relies on global pixel intensity histograms as features, Euclidean distance as a similarity measure, and k -means clustering; and **Approach B**, which employs deep CNN-based frame embeddings, cosine similarity, and a clustering method. A comprehensive trade-off analysis regarding robustness, computational efficiency, scalability, and feature impact follows.

1. Robustness, Semantics, and Feature Representation: Approach A uses color/intensity histograms, which are fundamentally orderless, spatially unaware descriptors. Two frames with identical global illumination and color distributions—but completely different spatial layouts or object placements—will yield zero distance, creating severe segmentation errors. Furthermore, Euclidean distance on raw histograms is sensitive to absolute magnitude differences and fails to capture perceptual similarity under varying illumination. Conversely, Approach B extracts high-level semantic embeddings from deep convolutional networks (e.g., pre-trained ResNet). These features encode hierarchical representations (edges, textures, shapes, and high-level semantic abstractions) that are inherently invariant to minor viewpoint changes, noise, and complex illumination shifts. It is crucial to note that these CNNs act purely as feature extractors for embedding generation, rather than performing explicit object detection or bounding-box localization.

2. Similarity Metrics and High-Dimensional Geometry: While Approach A uses Euclidean distance, Approach B relies on **cosine similarity**. In high-dimensional embedding spaces, vector norms are often uninformative or noisy, meaning two vectors can have vastly different magnitudes while representing the exact same semantic content. Cosine similarity computes the inner product normalized by the vectors' magnitudes ($\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$), making it strictly invariant to magnitude and focusing entirely on the vector direction. This aligns perfectly with the geometry of representation spaces.

3. Clustering Compatibility and Methodological Choice: A critical limitation in Approach B is the pairing of cosine similarity with k -means clustering. Standard k -means iteratively minimizes intra-cluster Euclidean distance, making it geometrically incompatible with cosine similarity. A more principled clustering strategy for Approach B would be:

- **DBSCAN:** It does not require specifying the number of clusters k a priori, natively supports arbitrary distance metrics (such as cosine distance), and effectively handles outliers (noisy frames).
- **Agglomerative Hierarchical Clustering:** Ideal for building structured taxonomies of video scenes using cosine distance.

4. Computational Efficiency and Dual Scalability:

- **Approach A** is extremely lightweight: histogram extraction runs in $O(N)$ time, and k -means converges rapidly on low-dimensional vectors. It is highly efficient and scales exceptionally well for resource-constrained edge devices (e.g., surveillance cameras or real-time CPU processing without GPUs).
- **Approach B** exhibits a dual nature of scalabilities:
 - *Computational scalability is low:* Forward CNN passes demand heavy hardware resources, making long-video processing expensive without GPUs or model compression (pruning, quantization).
 - *Generalization scalability is high:* A pre-trained CNN generalises effortlessly across diverse unseen video domains without requiring retraining or manual threshold tuning.

Scenarios where Approach A outperforms Approach B:

- **Abrupt Photometric Transitions:** Videos characterized by hard cuts, sudden flashes, or dramatic fades-to-black, where global intensity shifts are sufficient discriminators.
- **Resource-Constrained Systems:** Applications running on edge hardware, dashcams, or low-power CPUs where GPUs are unavailable and real-time performance is mandatory.
- **Unlabeled / Zero-Data Domains:** Scenarios where no pre-trained deep learning models are available or suitable for the specific video domain.

Scenarios where Approach B outperforms Approach A:

- **Semantic Content Changes with Constant Lighting:** Scenes where the overall illumination remains constant, but the underlying semantic content changes drastically (e.g., two different rooms illuminated by the same ambient light).
- **Complex Motion and Deformable Objects:** Videos featuring objects moving against static backgrounds, where high-level semantics are required to group coherent events rather than simple color statistics.

Impact of Feature Choice on Segmentation Quality: The choice of features fundamentally dictates the mathematical definition of “similarity” optimized by the system. Intensity histograms constrain similarity to global photometric distributions—a coarse and spatially insensitive proxy for visual content. CNN embeddings tie similarity to learned semantic proximity, which correlates significantly closer with human perceptual judgment. Consequently, the quality of segmentation (measured by boundary precision, reduction of false positives, and semantic coherence) is systematically higher with Approach B, traded off directly against computational tractability.



12.1 Homography limits and validity

A homography matrix H (3×3) maps points between two 2D images accurately if and only if at least one of the following two geometric conditions is satisfied:

- **Planar Scene:** The observed 3D scene points all lie on a single physical plane (e.g., a wall, a poster, the ground).

- **Pure Rotation:** The camera motion consists strictly of a pure rotation around its optical center ($t = 0$), regardless of the 3D structure of the scene.

What happens if the camera performs a pure translation (or a general motion with both rotation and translation) in a non-planar 3D scene? In this scenario, the **parallax effect** occurs. Points at different 3D depths project displacements in the image plane that are inversely proportional to their depth Z . Consequently, the geometric relationship between the two images can no longer be described by a single global homography, because the point correspondences depend on their individual spatial depths.

When parallax is present, homography estimation fails, and it is necessary to resort to **Epipolar Geometry**, estimating the **Fundamental Matrix** (F) or the **Essential Matrix** (E) to properly model the multi-view geometry.

❖ Mixed Questions

13.1 Zero-Padding vs. Mirror Padding in Sobel

Why does zero-padding generate a false high-gradient boundary along the entire image perimeter when followed by the Sobel operator?

Zero-padding adds a uniform layer of zeros around the image. When a derivative-based filter like Sobel is applied, the sharp intensity transition between the actual image content and the artificial border of zeros creates a strong, false high-gradient response along the perimeter. Differently, mirror (or reflect) padding mirrors the boundary pixel values symmetrically, smoothing the transition and preventing artificial gradient spikes.

13.2 Limits and Validity of Homography

Under what two specific geometric conditions does a Homography matrix H (3×3) accurately map points between two images? What happens if the camera undergoes a pure translation in a non-planar 3D scene?

A homography matrix H accurately maps points between two images if and only if at least one of two conditions is met: (1) the observed 3D scene is strictly planar, or (2) the camera motion is a pure rotation around its optical center ($t = 0$), regardless of the 3D structure. If the camera undergoes a pure translation (or general motion) in a non-planar 3D scene, the **parallax effect** occurs.

Points at different depths move by different amounts, making a single global homography invalid. In this scenario, one must resort to Epipolar Geometry using the Fundamental (F) or Essential (E) matrix.

13.3 Coarse-to-Fine Optical Flow and Inverse Warping

How do Gaussian pyramids, inverse warping, and residual flow solve the failure of Lucas-Kanade under large object motions?

Lucas-Kanade assumes small motion (e.g., < 1 pixel). Gaussian pyramids scale down the image resolution so that large motions (e.g., 20 pixels) at the original scale become small motions at the coarsest pyramid level. Starting from the coarsest level, the optical flow is computed, and **inverse warping** is used to align the second frame to the first without leaving gaps (avoiding forward warping holes). Then, a **residual flow** is computed for the remaining motion and propagated upward through the pyramid levels, accumulating the total optical flow.

13.4 Box Filter vs. Gaussian Filter in the Frequency Domain

Why does a Box filter produce ringing artifacts near sharp edges, while a Gaussian filter does not?

A Box filter is a rectangular function in the spatial domain, whose Fourier transform is a **Sinc function** in the frequency domain. The Sinc function exhibits prominent sidelobes that cause an abrupt and imperfect frequency cutoff, resulting in spatial ringing artifacts. Conversely, a Gaussian filter remains a Gaussian function in both spatial and frequency domains; it has no sidelobes and provides a smooth, clean attenuation of high frequencies without ringing.

13.5 Harris Corner Detector

What is the geometric intuition behind the Harris corner detector, and how are the eigenvalues of the Second Moment Matrix interpreted?

The Harris detector looks for local intensity variations by shifting a small window in all directions. It uses the Second Moment Matrix H (or M). By analyzing its eigenvalues (λ_1, λ_2):

- Both $\lambda_1, \lambda_2 \approx 0$ (Flat region): intensity is constant in all directions.

- One large, one near zero ($\lambda_1 \gg \lambda_2$ or vice versa) (Edge): intensity changes along only one direction.
- Both $\lambda_1, \lambda_2 \gg 0$ (Corner): intensity changes significantly in all directions.

13.6 Stereo Vision Baseline Trade-off

What is the engineering trade-off in stereo vision when choosing a very small baseline versus a very large baseline?

The baseline is the distance between the optical centers of the two cameras. A **narrow baseline** makes stereo matching much easier (high overlap, fewer occlusions), but depth estimation is very sensitive to noise, as small pixel disparity errors lead to huge errors in depth calculation (Z). A **wide baseline** provides high geometric precision for depth estimation, but makes stereo matching extremely difficult due to large perspective distortions and heavy occlusions.

13.7 The Aperture Problem

What is the aperture problem in optical flow estimation, and along which direction is motion ambiguous?

The aperture problem occurs when observing a moving edge through a small local window (aperture), making it impossible to determine the true 2D motion vector. Only the motion component perpendicular (normal) to the edge can be measured. The motion component **parallel to the edge** is completely ambiguous and cannot be reliably estimated because the intensity profile along the edge is uniform.

13.8 SIFT: Scale and Rotation Invariance

How does SIFT achieve scale invariance and rotation invariance?

Scale invariance is achieved by detecting keypoints across multiple scales using a scale-space representation based on the Difference of Gaussians (DoG) pyramids, locating stable extrema across adjacent scales. **Rotation invariance** is achieved by computing a local gradient orientation histogram in the neighborhood of each keypoint and assigning a canonical orientation corresponding to the highest peak of the histogram, so that descriptors are always aligned relative to this dominant direction.

13.9 Fundamental Matrix and its Rank

What is the Fundamental Matrix (F), and why does it have a rank of 2 ($\det(F) = 0$)?

The Fundamental Matrix F (3×3) is the algebraic representation of epipolar geometry for uncalibrated cameras, mapping a point in the first image to its corresponding epipolar line in the second image ($l' = Fx$). It has a **rank of 2** ($\det(F) = 0$) because it maps a 2D point onto a 1D line (epipolar line). Mathematically, it is defined as $F = K'^{-T}[t]_{\times}RK^{-1}$, where the essential matrix component $[t]_{\times}$ is a skew-symmetric matrix of rank 2, which forces F to also be singular (rank 2).

13.10 Epipoles and Epipolar Lines

What are epipoles and epipolar lines, and how do they simplify stereo matching?

An **epipole** (e, e') is the intersection of the baseline (the line connecting the two camera centers) with the respective image planes (equivalently, the projection of one camera center onto the other image plane). An **epipolar line** is the intersection of the epipolar plane (defined by 3D point X and the two camera centers) with the image plane. The epipolar constraint simplifies stereo matching from a 2D search problem to a 1D search problem: given a point x in the first image, its corresponding point x' in the second image must lie entirely along its corresponding epipolar line l' .

13.11 RANSAC vs. Least Squares

Why do Least Squares fail in the presence of high outlier ratios, and how does RANSAC solve this?

Standard Least Squares minimizes the sum of squared residuals, which causes even a small percentage of severe outliers (e.g., 40% false matches) to drastically pull and distort the estimated model parameters. **RANSAC (Random Sample Consensus)** solves this by iteratively sampling the minimum subset of points required to estimate the model, computing the transformation, and counting how many total points agree with it (the consensus set / inliers). By keeping the model with the largest consensus set over many iterations, it completely isolates and discards the outliers.

13.12 Mean vs. Median Filter for Salt-and-Pepper Noise

Why does a Mean filter blur salt-and-pepper noise while a Median filter completely eliminates it?

A Mean filter is a linear filter that averages all pixel values in a neighborhood. Since salt-and-pepper noise consists of extreme, isolated impulse values (maximum or minimum brightness), averaging them smears the noise values across neighboring pixels, causing blurring. A Median filter is a non-linear filter that replaces the pixel value with the statistical median of its neighborhood. Extreme noise impulses fall at the extreme ends of the sorted neighborhood values and are automatically ignored, completely removing the noise without blurring edges.

13.13 Phase Swapping Experiment in Fourier Domain

In a phase-swapping experiment between Image A and Image B, which image is visually recognizable in the IFFT output and why?

Image B (the one whose **Phase** was used) will be visually recognizable. While the Amplitude spectrum contains the overall energy distribution and frequency magnitudes, the **Phase spectrum** contains the precise spatial structure, alignment, and edge locations of the image. The human visual system relies overwhelmingly on phase information to perceive shapes and objects.

13.14 Background Subtraction in Outdoor Environments

What traditional (non-Deep Learning) approach is used for background subtraction in outdoor scenes with changing illumination and moving shadows?

Simple frame differencing fails due to lighting variations and shadows. Instead, traditional computer vision uses statistical background modeling such as **Gaussian Mixture Models (GMM)**, where each pixel is modeled over time by a mixture of Gaussians to account for multi-modal backgrounds (e.g., swaying tree leaves, flickering light). Alternatively, color spaces like HSV are used, separating chromaticity from intensity (V) to make the system more robust to illumination shifts and shadows.

13.15

13.16 Tracking Failures: Template Matching vs. Optical Flow

Consider two pedestrian tracking systems in a video: System A uses template matching (SSD against a fixed patch from $t = 0$), and System B uses optical flow (iteratively propagating position). Both work for 10 seconds, but fail after 30 seconds. Explain the failure mechanisms and design a non-Deep Learning hybrid system.

1. System A Failure Mechanism (Template Matching): System A does not fail due to keypoint deformation (since template matching operates on raw pixel patches via SSD/NCC rather than discrete feature keypoints). Instead, it fails due to **template staleness**. Because the reference template is statically fixed to frame $t = 0$ and never updated, it cannot represent a non-rigidly deforming, evolving object. Over 30 seconds, the pedestrian's pose (limb articulation during walking), scale (approaching/receding), illumination (shadows, direct sunlight), and viewpoint change completely. The static patch no longer resembles the target, causing SSD to find an incorrect local minimum. This failure is structurally inevitable for any non-rigid or scaling object.

2. System B Failure Mechanism (Optical Flow): System B fails due to **temporal error accumulation (drift)** combined with structural vulnerabilities. Lucas-Kanade assumes *small motion* between consecutive frames, introducing a small estimation error ϵ per frame. This error accumulates linearly over time ($T \cdot \epsilon$), resulting in a significant spatial offset after 900 frames (30 seconds at 30 fps). Furthermore, System B fails inevitably during **occlusions** (partial or total): because optical flow cannot estimate motion on invisible pixels, the propagation breaks down immediately and cannot recover independently.

3. Non-Deep Learning Hybrid System Design: To overcome these limitations without Deep Learning, we can design a robust coarse-to-fine hybrid tracker combining both approaches with safety mechanisms:

- 1. Motion Propagation via Optical Flow (LK):** Use optical flow on pixels within the bounding box to estimate local motion and predict the target's rough position in frame $t + 1$. This handles continuous motion efficiently.
- 2. Local Refinement via Template Matching (NCC):** Perform a localized search (using Normalized Cross-Correlation) around the predicted position to correct sub-pixel drift and small accumulation errors.
- 3. Adaptive Template Update:** To prevent template staleness (System A's

flaw), if the NCC confidence score remains high but shows minor appearance shifts, the reference template is **adaptively updated** with the current patch at a controlled, conservative rate.

4. **Confidence Thresholding & Loss of Target Declaration:** If the NCC confidence score drops below a strict threshold (due to severe occlusions or tracking failure), the system suspends template updates and declares a **loss of target** rather than blindly propagating corrupted tracking windows.

13.17

13.18 Industrial Conveyor Belt Anomaly Detection

Design an automated industrial anomaly detection system for a moving conveyor belt using traditional computer vision. Evaluate a colleague's naïve proposal of using direct pixel-by-pixel background subtraction against a static empty-belt reference image.

a) Valid Conditions for the Naïve Approach: This direct subtraction approach works correctly and is acceptable only if strict conditions are met: (1) The conveyor belt must be stationary (or perfectly synchronized) during image capture; (2) The object must stop at the exact same identical position every time; and (3) The anomaly or object must exhibit a high-contrast difference in color/intensity relative to the belt. Under these rigid constraints, simple thresholding is sufficient because spatial coordinates remain perfectly aligned.

b) Three Fundamental Failure Modes in Dynamic Conditions: When the belt moves continuously in a real industrial setting, the naïve approach fails due to:

1. **Non-deterministic Motion Blur:** If the belt moves continuously, the textured pattern of the rubber blurs. Since belt speed varies slightly between cycles, the blur pattern changes, causing systematic false positives during subtraction.
2. **Mechanical Vibrations:** Industrial motors induce micro-vibrations in the camera mounting, causing subtle frame-to-frame misalignments (spatial jitter) between the current frame and the static reference.
3. **Object Misalignment (Ghosting Artifacts):** Even identical objects slip or shift slightly by 2–3 pixels. Subtraction creates a severe double-edge “ghosting” artifact around the object boundaries, leading to false anomaly detections.

4. (Bonus) **Accumulation of Dirt and Wear:** Over time, dust, grease, and scratches accumulate on the belt. A static reference image captured at $t = 0$ becomes obsolete, generating widespread false positives.

c) Robust Traditional Computer Vision Improvements: To maintain computational simplicity while fixing these flaws without Deep Learning:

1. **For Motion Blur:** Use an **encoder trigger** synchronized with the belt motor (or a strobe flash) to freeze motion at the exact same inspection window, eliminating motion blur at the root.
2. **For Mechanical Vibrations:** Apply a fast **rigid image registration** (translation and rotation via frequency-domain cross-correlation) to stabilize and realign the frame with the reference before subtraction.
3. **For Object Slippage/Positioning:** Use localized feature alignment or an optimized template matching/homography to warp the inspected object into standard alignment, preventing boundary ghosting.
4. **For Belt Wear/Dirt:** Implement a **periodic reference update** algorithm (running background update during empty belt intervals) to dynamically adapt the reference image.

13.19

13.20 Panoramic Stitching Failures: Corridor vs. Open Piazza

A panoramic stitching pipeline (SIFT + Ratio Test + RANSAC + Global Homography) works perfectly in a closed corridor, but fails with ghosting and misalignment in an open piazza with buildings at different depths, despite abundant matches and correct RANSAC convergence. Explain the geometric causes and how to solve it.

a) Geometric Difference (Corridor vs. Piazza): In a closed corridor, all scene points lie approximately at the same depth and on coplanar surfaces. A homography mathematically models this exactly, as it is valid for planar scenes or pure camera rotation. In an open piazza, buildings and structures exist at radically different depths. When the camera rotates, foreground objects undergo a larger apparent displacement than distant background objects. This differential motion is **parallax**, which violates the fundamental assumptions of a single global homography.

b) Why RANSAC Converges but the Result is Wrong: RANSAC successfully finds the homography that maximizes the number of inliers, but it is optimizing over a **misspecified model**. Since no single global homography can simultaneously map points at varying depths, RANSAC converges to a least-squares compromise—typically aligning distant background points while failing on foreground elements. It is analogous to fitting a straight line to curved data: the algorithm optimizes correctly, but the underlying geometric model is insufficient for a multi-depth scene.

c) Solution (Geometric Correction + Blending): To solve this properly without relying on naive blending (like feathering, which only masks geometric errors):

1. **Correct Geometric Model (Cylindrical Projection):** Instead of warping flat image planes via homography, project images onto a cylindrical or spherical surface. Under cylindrical coordinates, pure camera rotation reduces to a simple horizontal translation, inherently eliminating background parallax.
2. **Multi-Band Blending:** For any residual parallax in foreground objects, apply a Laplacian pyramid blending technique. This decomposes images into frequency bands, smoothing low frequencies (illumination/color) over wide zones while blending high frequencies (edges/details) locally, ensuring sharp transitions without ghosting.

13.21

13.22 Document Rectification: Passports vs. Receipts

A document rectification pipeline (SIFT + RANSAC + Homography) works well on passports and IDs, but systematically fails on supermarket receipts, despite them being flat rectangles. Explain why SIFT fails, propose a traditional alternative for corner detection, and explain how to estimate and apply the homography.

a) Why SIFT Fails on Supermarket Receipts: Supermarket receipts lack distinctive, unique local texture. They consist of repetitive rows of text, identical characters, and large white spaces, which create severe descriptor ambiguity for SIFT. Furthermore, thermal paper lacks the rich graphical features, holograms, complex background patterns, and sharp internal gradients found in passports or IDs. Consequently, SIFT fails to find reliable, unambiguous matches across views.

b) Traditional Alternative for Corner Detection (No SIFT): Instead of feature matching, we use a classic image processing pipeline:

1. **Binarization:** Apply Otsu's thresholding to separate the receipt from the background table.
2. **Edge Detection & Contour Extraction:** Use Canny edge detection followed by contour tracing to find closed boundaries.
3. **Polygon Approximation:** Apply the Douglas-Peucker algorithm to simplify the largest closed contour into a quadrilateral polygon guaranteed to have exactly 4 vertices. The geometric intersections of these four lines yield the 4 precise corner coordinates.

c) Estimating and Applying the Homography:

1. **Target Coordinates:** Determine the target dimensions (width and height) by computing the maximum Euclidean distances between the detected corners. Define the destination points as a clean, frontal rectangle: $(0, 0)$, $(\text{width}, 0)$, $(\text{width}, \text{height})$, and $(0, \text{height})$.
2. **Homography Estimation:** An 8-DOF homography matrix H requires 4 point correspondences (yielding 8 linear equations). Construct the system $Ah = 0$ and solve it using Singular Value Decomposition (SVD), taking the right-singular vector corresponding to the smallest singular value.
3. **Rectification via Inverse Warping:** Apply the inverse of the estimated homography to map destination pixel coordinates back into the source image, using bilinear interpolation. This ensures every pixel in the rectified output is filled, preventing interpolation gaps.

13.23

13.24 Document Rectification: Passports vs. Receipts

A document rectification pipeline (SIFT + RANSAC + Homography) works well on passports and IDs, but systematically fails on supermarket receipts, despite them being flat rectangles. Explain why SIFT fails, propose a traditional alternative for corner detection, and explain how to estimate and apply the homography.

a) Why SIFT Fails on Supermarket Receipts: Supermarket receipts lack distinctive, unique local texture. They consist of repetitive rows of text, uniform

thermal printing, and large white spaces, which create severe descriptor ambiguity for SIFT. Unlike passports or IDs—which contain rich graphical elements, color photographs, holograms, and sharp multi-directional gradients—receipts offer no robust local keypoints that can be matched reliably across varying viewpoints.

b) Traditional Alternative for Corner Detection (No SIFT): Instead of feature matching, we use a robust bottom-up image processing pipeline:

1. **Binarization & Edge Detection:** Apply Otsu’s thresholding (noting that under uncontrolled conditions with similar background tones, contrast enhancement or adaptive thresholds may be required) followed by Canny edge detection to highlight structural boundaries.
2. **Contour Extraction & Morphology:** Extract connected components using contour finding algorithms (`findContours`). Apply morphological closing operations (dilation followed by erosion) to bridge any small gaps or broken edges in the receipt boundary.
3. **Largest Contour Selection & Polygon Approximation:** Select the largest closed contour by area (corresponding to the document). Apply the Douglas-Peucker polygon approximation algorithm to reduce the contour vertices down to a quadrilateral with exactly 4 vertices, representing the receipt’s corners.

c) Estimating and Applying the Homography:

1. **Target Rectangle Estimation:** Compute the real-world proportions by taking the maximum Euclidean distances between opposite corners (e.g., width as $\max(d_{top}, d_{bottom})$ and height as $\max(d_{left}, d_{right})$). Define the destination coordinates as a clean frontal rectangle from $(0, 0)$ to $(width, height)$.
2. **Homography Estimation via SVD:** With 4 point correspondences, construct the direct linear transformation system $Ah = 0$ (yielding 8 equations for 8 DOFs). Solve it using Singular Value Decomposition (SVD), where the estimated homography vector corresponds to the right-singular vector associated with the smallest singular value.
3. **Rectification via Inverse Warping:** Apply **inverse warping** (mapping destination coordinates back to the source via H^{-1}) combined with bilinear interpolation. Inverse warping is strictly required because forward mapping leaves unmapped destination pixels (holes/gaps), whereas inverse

mapping guarantees that every pixel in the rectified output frame is evaluated and filled.

13.25

13.26 Stereo Vision Failure Scenarios: A, B, and C

A calibrated and rectified stereo vision system works in a lab, but degrades in three specific scenarios: A (white uniform walls), B (outdoor scene with a window), and C (two objects at different depths with a sharp boundary). Explain the failure mechanism, the traditional solution, and the violated block matching assumption for each.

Scenario A: Uniform White Walls (Aperture / Textureless Region)

- **Failure Mechanism:** The smooth white wall lacks distinctive high-frequency texture or intensity gradients. Consequently, local pixel windows look identical, producing a flat, ambiguous matching cost curve (SSD) with no distinct global minimum, resulting in large unassigned grey areas.
- **Violated Assumption:** *Texture Sufficiency* (the assumption that every local patch contains sufficient intensity variation to be uniquely matched).
- **Traditional Solution:** Project an artificial structured light pattern (e.g., an infrared dot pattern or laser texture) onto textureless surfaces to force high-frequency features.

Scenario B: Window with Reflections (Specular/Non-Lambertian Surface)

- **Failure Mechanism:** Glass acts as a semi-reflective mirror. Because the left and right cameras view the window from different angles, they capture completely different reflections (e.g., indoor room vs. outdoor sky), violating photometric consistency.
- **Violated Assumption:** *Brightness Constancy / Photometric Consistency* (the assumption that a physical point reflects the same light intensity into both cameras).
- **Traditional Solution:** Use cross-polarization (placing orthogonal polarizing filters on the cameras and light sources) to eliminate specular reflections, or apply disparity confidence masks to filter out regions with high left-right photometric inconsistency.

Scenario C: Sharp Boundary Between Near and Far Objects (Fattening Effect)

- **Failure Mechanism:** When the block matching window straddles a sharp depth discontinuity, it simultaneously captures pixels belonging to two different depth planes. No single patch in the right image can match this mixed combination, causing a smeared cost function, erroneous disparity values, and the **fattening effect** (foreground boundaries “bleeding” onto the background).
- **Violated Assumption:** *Fronto-Parallel Surface Assumption* (the assumption that all pixels within the matching window lie on a single continuous flat surface at constant depth).
- **Traditional Solution:** Employ adaptive support weights (adaptive windows that shrink or adjust near edges) or Global/Semi-Global Matching (SGM) with regularization terms that penalize abrupt, un-supported disparity jumps.

13.27

13.28 Diagnosing Low Inlier Percentage in RANSAC

A SIFT + RANSAC homography pipeline yields an inlier ratio of 29% (87 out of 300 matches). A colleague suggests increasing RANSAC iterations N to fix it. Why is the colleague wrong, what are the photometric and geometric causes, and how would you diagnose and solve them?

Vincoli della domanda: A pipeline with SIFT, RANSAC, and Homography successfully converges despite a low inlier ratio, indicating a structural issue in the data rather than an algorithmic failure of RANSAC.

a) Why the Colleague is Wrong: The colleague is wrong because N (number of iterations) only guarantees the statistical probability p of randomly sampling at least one outlier-free set of minimum points ($s = 4$). If the inlier ratio is low due to structural data issues (bad matches or model misspecification), running RANSAC longer will not magically turn outliers into inliers; it will just test the same flawed pool of matches more times.

b) Photometric and Geometric Causes:

- **Photometric Cause:** Reflections, specularities (non-Lambertian surfaces like glass windows on urban buildings), or severe illumination changes between

views that cause SIFT descriptors to fail or fail Lowe’s ratio test, resulting in a high rate of incorrect initial matches.

- **Geometric Cause:** A violation of the homography assumptions due to strong 3D **parallax** (buildings at vastly different depths in an urban scene). A single global homography cannot model multi-depth structures.

c) Diagnosis and Solutions:

- **Diagnostic Method:** Visualize the spatial distribution of RANSAC inliers on the image. If inliers are densely clustered only on a single flat surface (e.g., one building facade) while the rest of the scene is rejected, the issue is **geometric (parallax)**. If inliers are sparsely scattered or absent on reflective/shadowed areas, the issue is **photometric**.
- **Solutions:**
 - For *Geometric issues*, abandon the global homography model and use multi-view geometry (Epipolar geometry / Fundamental Matrix) or piecewise planar stitching.
 - For *Photometric issues*, apply preprocessing like CLAHE (Contrast Limited Adaptive Histogram Equalization) or use robust feature descriptors less sensitive to specularities.

13.29

13.30 Calibrated vs. Uncalibrated Stereo Systems and Zhang’s Calibration

Two identical stereo hardware setups exist: System A has calibrated cameras (K, R, t known), and System B is uncalibrated. A colleague claims that because they share identical hardware and see the same scene, their 3D reconstruction will be similar. Why is the colleague wrong, what can and cannot System B do, and how does Zhang’s method estimate K ?

Vincoli della domanda: The colleague confuses identical hardware with identical geometric information; the core issue is not what the cameras see, but what can be mathematically computed without K .

a) Why the Colleague is Wrong — The Role of K : The camera projection matrix $P = K[R \mid t]$ maps a 3D world point X_w to a 2D image point x in pixels: $\lambda x = K[R \mid t]X_w$. The intrinsic matrix K encodes optical properties (focal lengths f_x, f_y , principal point c_x, c_y). It acts as the mathematical bridge converting

metric 3D coordinates into pixel coordinates. The colleague is wrong because identical hardware does not imply identical K (manufacturing tolerances vary focal lengths and optical centers). Without K , true metric 3D reconstruction is fundamentally impossible, regardless of identical hardware.

b) What System B Can and Cannot Do: System B without K is not completely blind, but faces strict mathematical boundaries:

- **Still possible without K :** Extract SIFT/SURF keypoints and descriptors, perform feature matching, estimate the Fundamental matrix F via the normalized 8-point algorithm, and perform epipolar rectification.
- **Structurally blocked without K :** Cannot convert F to the Essential matrix E (since $E = K'^T F K$), cannot recover metric rotation R and translation t , cannot triangulate in metric 3D coordinates (yielding only a *projective reconstruction* up to an unknown projective transformation), and cannot measure real-world distances.

c) Zhang's Method for Estimating K : Zhang's method estimates K from a planar calibration target (checkerboard) without requiring 3D world points:

1. **Capture Multiple Views:** Photograph the planar checkerboard ($Z = 0$ in its local frame) from $N \geq 3$ different poses.
2. **Estimate Homographies:** For each view, estimate the homography $H_i = K[r_1 \ r_2 \ t]$ linking board points to image points using DLT.
3. **Extract Constraints on K :** Each homography provides 2 constraints on K via the orthonormality of rotation columns ($r_1^T r_2 = 0$ and $\|r_1\| = \|r_2\|$), expressed through the Image of the Absolute Conic (IAC) $\omega = K^{-T} K^{-1}$.
4. **Solve for K :** With $N \geq 3$ views, we get ≥ 6 constraints, sufficient to solve for K (5 DoF) linearly via SVD, recovering K through Cholesky decomposition of ω^{-1} .
5. **Refinement:** Minimize total reprojection error via Bundle Adjustment.